Prasiddha Thapa

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Education

Howard University – COAS Honors Program (Presidential Scholar) Expected graduation: 2030
B.S. in Mathematics; intended second major in Data Science

Current coursework: Proof and Problem Solving; Introduction to Data Science; Mathematics Research I

Selected Honors

67th International Mathematical Olympiad — Honorable Mention 2026
Represented Nepal; joint-highest score on the national team.

Nepal Mathematical Olympiad — Top 3 nationally 2026
Placed in the top 3 through a qualification system with 5,000+ participants.

4th International Mathematics Summer Camp — Bronze Medal 2026
Individual competition; highest-scoring participant from Nepal. Beijing, China.

National Math League — 1st Place, Team 2025

Research & Computational Projects

Temporal Evidence Accumulation in Inhibitory/Excitatory Spiking Neural Networks 2026
Research supervised by Professor Christopher Kim, Howard University

- Implemented and trained a recurrent spiking neural network in PyTorch with an **80/20 excitatory/inhibitory split neuron population**, leaky integrate-and-fire dynamics, Dale-constrained connectivity, and surrogate gradients to integrate noisy evidence over time.
- Evaluated circuit function through ablation experiments: **86.8% accuracy** for the full network versus **63.9% without recurrence** and **48.1% without inhibition** on a synthetic two-choice evidence-accumulation task.
- Characterized decisions across evidence strengths and durations; accuracy increased from approximately **69% at 100 ms to 93% at 800 ms**. Analyzed excitatory/inhibitory currents to connect circuit dynamics with behavioral performance.

Household Pet-Readiness Modeling 2024
International Mathematical Modeling Challenge — Honorable Mention, Team

- Led mathematical modeling and data collection; implemented a Python readiness index combining financial, healthcare, and environmental factors using normalization and weighted scoring.
- Applied the model to demographic and economic data from the United States, Germany, and Canada; adapted its assumptions for different pet species and multi-pet households.

Mathematical Teaching & Leadership

Problem Setter & Tutor — ClampHook / Private Tutoring

- Created and edited mathematics question sets in LaTeX; prepared problem sheets and lesson plans, and tutored two students in mathematics and science.

Writer — Olympiad Mathematics Handouts

- Co-authored and typeset preparation material on algebraic inequalities and number theory to prepare handouts in LaTeX under Prajit Adhikari, a former IMO participant, for the math olympiad participants.

Founder & Captain — Mathematics Olympiad Team

- Led preparation sessions and captained teams representing St. Xavier's College; also mentored beginners through Nepal's national mathematics Discord community.

Technical & Mathematical Skills

Programming: Python, PyTorch, NumPy, scikit-learn, Matplotlib, LaTeX
Methods: Monte Carlo simulation; familiarity with simulated annealing; proof writing